

**Jawaharlal Nehru Govt. Engineering College Sundernagar, Mandi (H.P.)**

Course Name: **Basic Electrical Engg.** Course Code: EE-111 Semester: 1st

Examination: B.Tech

Scheme: CBS

Branch: CSE, ECE, CE

**Question Paper for 1st Periodical Examinations (Oct, 2024)**

Time: Two Hours

Max. Marks: 50

**Course Outcomes:-**

CO101.1. Recalling Basic Electrical and magnetic laws

CO101.2 Remembering Steady State analysis of DC circuits

CO104.3 Understanding AC Single phase & Three phase Circuits

CO104.4 Solving complex AC, DC circuits

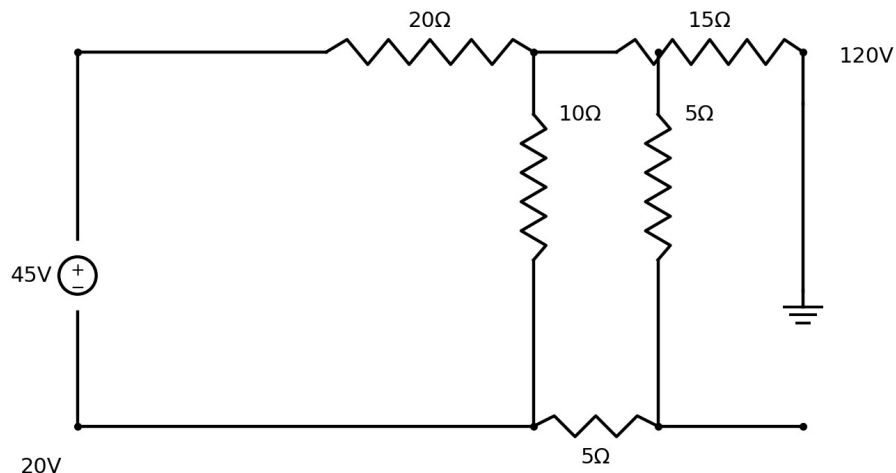
CO104.5 Explaining various concepts of Electrical machines

Q1. (CO101.1) Derive Expression for star to delta and delta to star transformation? **(10)**

Q2. (CO101.3) (a) What is power factor? What is its significance? What are different methods to find power factor? **(5)**

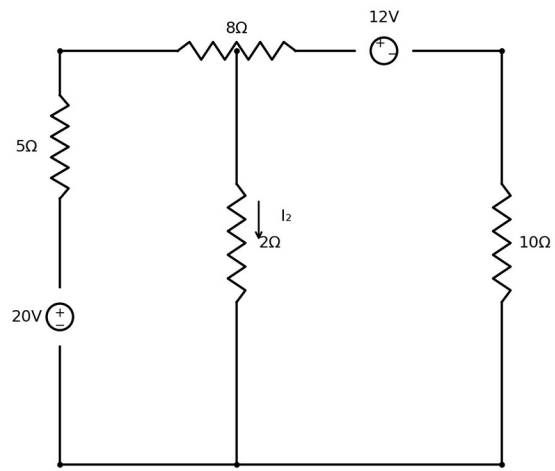
(b) State and explain Thevenin's Theorem? **(5)**

Q3. (CO101.1) Find current flowing in 15 ohm resistor using nodal analysis? **(10)**



Q4. (CO101.2) Derive the expressions for Transient Response of series RC circuit having DC Excitation? **(10)**

Q5. (CO101.4) Find the current flowing in the 10 ohm resistor using Norton's theorem. Also draw Norton's Equivalent circuit? **(10)**



(Course Advisor: Manavi Sharma)